



Introduction to Database Systems

SPRING 2026

Introduction

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Course Books

- ▶ **Text Book:**

- ▶ Ramez Elmasri, Fundamentals of Database Systems (7th /8th Edition)
- ▶ Link for Database books:
 - ▶ https://onlinepgc-my.sharepoint.com/:f/g/personal/sadaf_baloch_ucp_edu_pk/EurZuqbVjzRLsZ2gVhpslfcBLN4Dbk-X6R6GJtWBkShHkQ?e=6nIHdh

- ▶ **Marks Division:**

- ▶ Assignments (15%)
- ▶ Quizzes (15%)
- ▶ CP (10%)
- ▶ Mid Term (20%)
- ▶ Final Term (40%)

Announcements & Plagiarism

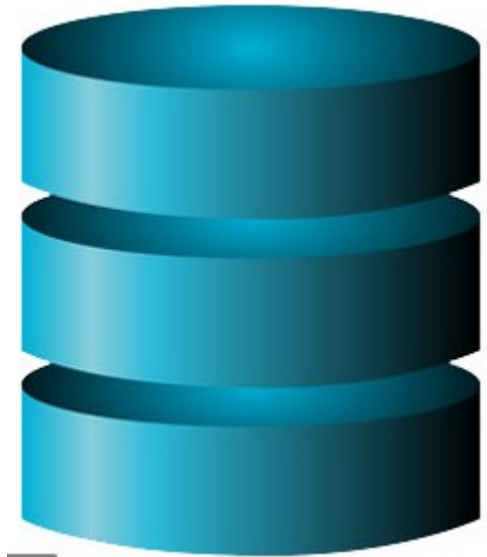
- ▶ Make sure you check your email and announcement on portal regularly. I keep on posting announcements for the deadlines.
- ▶ Plagiarism Policy will be strictly applied.
- ▶ No retakes

Lecture 1

INTRODUCTION TO DATABASES

Database

- ▶ **Database**
 - ▶ The collection of data
 - ▶ Collection of related data



Properties of Databases

A database has the following implicit properties:

- ▶ A database represents some aspect of the real world, sometimes called the **mini world**. Changes to the miniworld are reflected in the database.
- ▶ A database is designed, built, and populated with data for a specific purpose. It has an intended group of users and some preconceived applications in which these users are interested.

Some examples

Emergence of web interfaces provided direct database interaction. Data on website is directly fetched from the database.

Examples:

- ▶ **Banking:**
 - ▶ For customer information, accounts, loans and banking transactions etc.
- ▶ **Airlines:**
 - ▶ For reservations and schedule information.
- ▶ **Universities:**
 - ▶ For student information, course registration and grades.
- ▶ An example of a large commercial database is Amazon.com

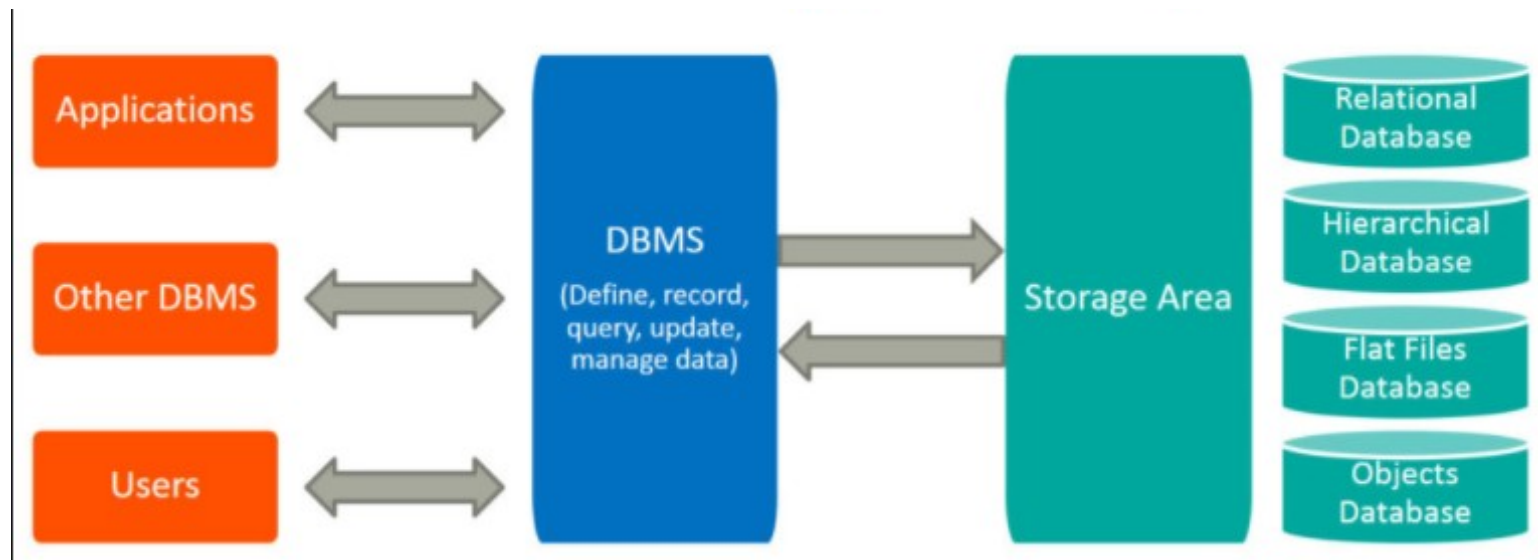
DBMS

► Database Management System

- Collection of interrelated data and a set of programs to access those data.

► Goal of DBMS

- To provide a way to store and retrieve database information that is both convenient and efficient.

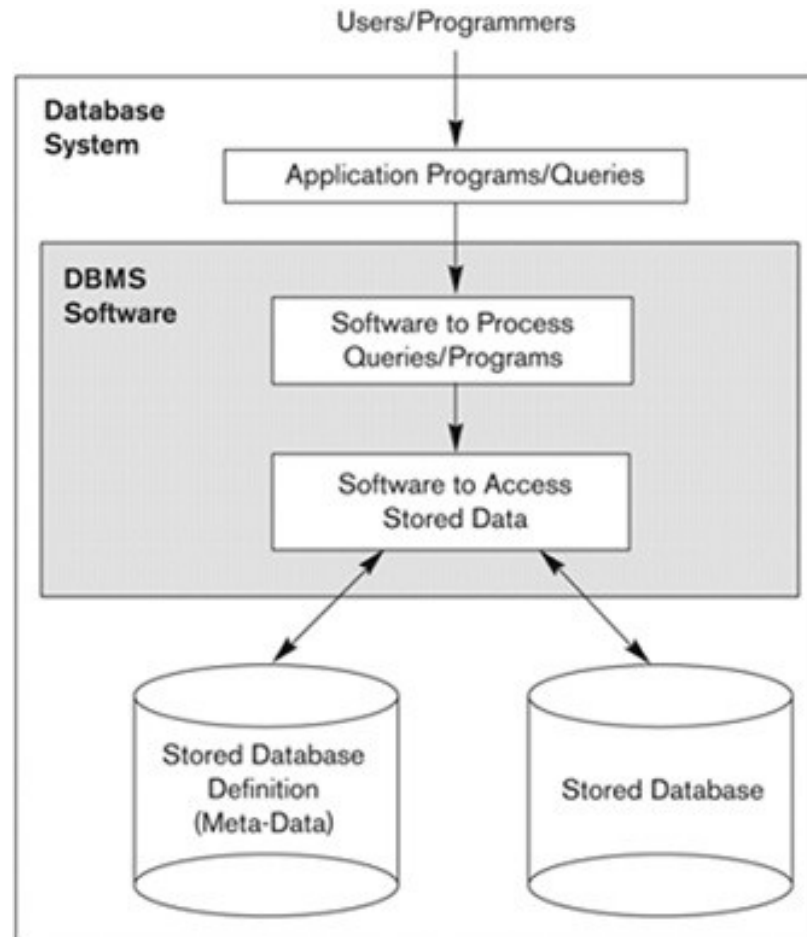


DBMS Functionality (Why to use DBMS)

- ▶ **Define** a database in terms of data types, structures and constraints
- ▶ **Construct** or Load the Database on a secondary storage medium
- ▶ **Manipulating** the database by querying, generating reports, insertions, deletions and modifications to its content
- ▶ **Sharing** by a set of users and programs but keeping the data consistent at the same time.
- ▶ **Security** measures to prevent unauthorized access.
- ▶ **Presentation** and Visualization of data

A simplified database system environment

Slide
1-11



Some points to ponder

- ▶ Why are we using DBMS?
- ▶ What we have been using till now?
- ▶ Why are we moving from traditional file system to database system?

Issues in filling
system.

Activity Time

- ▶ Lets see an excel sheet holding some records.
- ▶ Please see if you find out any problems in the given excel sheet.
- ▶ Problems related to :
 - ▶ The current state of the file.
 - ▶ If you want to add some other rows.
 - ▶ If you want to delete some existing rows.
 - ▶ If you want to update some information in the existing rows.

File

Home

Insert

Page Layout

Formulas

Data

Review

View

Team



Paste



Format Painter



Clipboard

Times New Roman

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StylesInsert
CellsDelete
CellsFormat
Cells

AutoSum



Fill



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Sort &
Filter

Select

Editing

A1

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Student Id

	A	B	C	D	E	F	G	H	I	J	K	L	M
	<i>Student Id</i>	<i>Student FirstName</i>	<i>Student LastName</i>	<i>Student Address</i>	<i>Student ContactNo</i>	<i>Course Id</i>	<i>Course Name</i>	<i>Course Grades</i>	<i>Instructor Id</i>	<i>Instructor Name</i>	<i>Instructor Address</i>	<i>Instructor ContactNo</i>	
1													
2	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV101	Linear Algebra	A	I_01	Paul Koch	9012, rue Royale	34.56.78.90	
3	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV204	Visual Basic	B	I_03	Michael Allen	Kirchgasse 9012	(171) 789-0123	
4	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV301	Wellness	A	I_02	Mark Hassall	Fauntleroy Circus 4567	(1) 890-1234	
5	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV404	Political Science	C	I_10	John Peoples	Cerrito 3456	(5) 456-7890	
	S1	Jerry	Robinson	Obere Str. 0123	030-3456789	UV501	Network Security	A	I_13	Tom Higgin	Hauptstr. 0123	(11) 012-3456	
6													
7	S2	John	Walker	Avda. de la 5678	(5) 789-0123	UV110	Calculus-2	C	I_41	Venky Krishnan	6789 Baker Blvd.	(503) 555-0122	
8	S2	John	Walker	Avda. de la 5678	(5) 789-0123	UV208	MicroComputer	B	I_23	Shabalin Rostisl	C/ Romero, 1234	(95) 901 23 45	
9	S2	John	Walker	Avda. de la 5678	(5) 789-0123	UV303	Calculus	C	I_12	Jacek Jelitto	Hauptstr. 0123	0452-678901	
10	S2	John	Walker	Avda. de la 5678	(5) 789-0123	UV404	Political Science	A	I_10	John Peoples	Cerrito 3456	(5) 456-7890	
	S2	John	Walker	Avda. de la 5678	(5) 789-0123	UV501	Network Security	A	I_13	Tom Higgin	Hauptstr. 0123	(11) 012-3456	
11													
12	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV101	Linear Algebra	A	I_01	Koch, Paul	9012, rue Royale	34.56.78.92	
13	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV204	Visual Basic	B	I_03	Michael Allen	Kirchgasse 9012	(171) 789-0123	
14	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV301	Wellness	A	I_02	Mark Hassall	Fauntleroy Circus 4567	(1) 890-1234	
15	S3	Mary	Mosis	Mataderos 7890	(5) 123-4567	UV407	Geography	B	I_20	Tianna Jones	Walserweg 4567	(604) 901-2345	

1. Redundancy

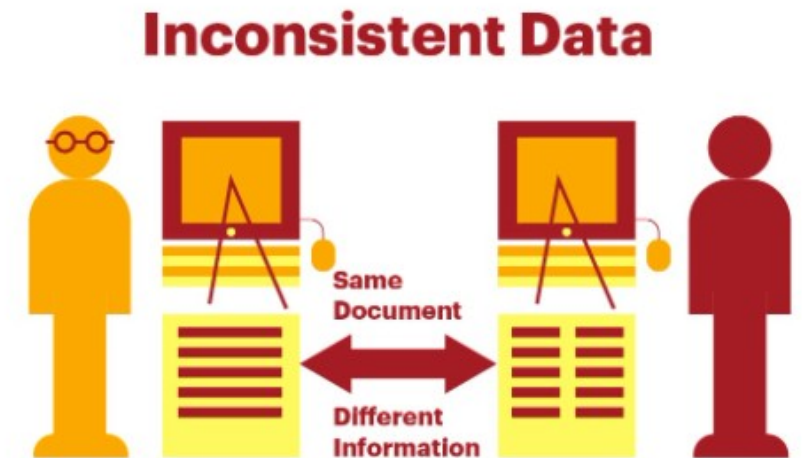
- ▶ Duplicates of data
- ▶ Same piece of information present at multiple locations

Student_ID	Name	Contact	College	Course	Rank
100	Himanshu	7300934851	GEU	Btech	1
101	Ankit	7900734858	GEU	Btech	1
102	Aysuh	7300936759	GEU	Btech	1
103	Ravi	7300901556	GEU	Btech	1

2. Inconsistency

- ▶ Different information related to same entity in different/same files.
- ▶ Data can be inconsistent if:
 - ▶ Changes are made to one place and not in another
 - ▶ There is a typo
 - ▶ Data is present in different formats
- ▶ Data redundancy can cause data inconsistency

student_id	student name	teacher_id	teacher_name
1	A	1	P
2	B	1	P
3	C	2	Q
4	D	2	R



3. Data Isolation & Data Mapping

- ▶ Data is scattered in various files and may be in different formats.
 - ▶ Writing new programs for different files may be difficult.
- ▶ No link between the files
 - ▶ Changes in one cannot be reflected in another

IID	LNAME	FNAME	RANK	SALARY	DEPT
201	Gauss	Fred	Prof	55000	Math
301	Joyce	James	Assoc Prof	23000	English

CID	TITLE	SEMESTER	UNITS	LOCATION	TIME	INSTRUCTOR
MATH23	Calculus	Fall 2000	3	MSRI1	MWF 1-2	201
ENG1A	Literature	Fall 2000	5	DWIN2	TTh 3-4	301

4. Integrity Problems

- ▶ Age must be under 60
 - ▶ Account balance must not be negative
 - ▶ Balance must be 5000 or greater. If this rule changes, changes would be done at every check.
-
- ▶ How to handle all these?

5. Atomicity

- ▶ For example: We are transferring some amount to another account and there is a power outage. What will happen?

Metadata/Data Dictionary/Catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Design phases of databases

How is a house constructed?

- ▶ If you want to build a house what steps will you follow?
 - ▶ We will give our requirements to the architect.
 - ▶ He will show us some paperwork i.e. a map or a 3d model.
 - ▶ Then the foundations are laid and grey structure is built according to the preferred material.
 - ▶ We then finalize about the finishing and furnishing.

Design Phases of Database

1. Requirement Definition and analysis

- ▶ Design of a new application for an existing database

OR

- ▶ Design of a new database

2. Conceptual Design

- These requirements are documented in detail to form a conceptual design
 - Conceptual design can be represented and manipulated using some computerized tool so that it can easily be maintained, modified and transformed to database implementation.

Example: Entity Relationship Model

Design Phases of Database

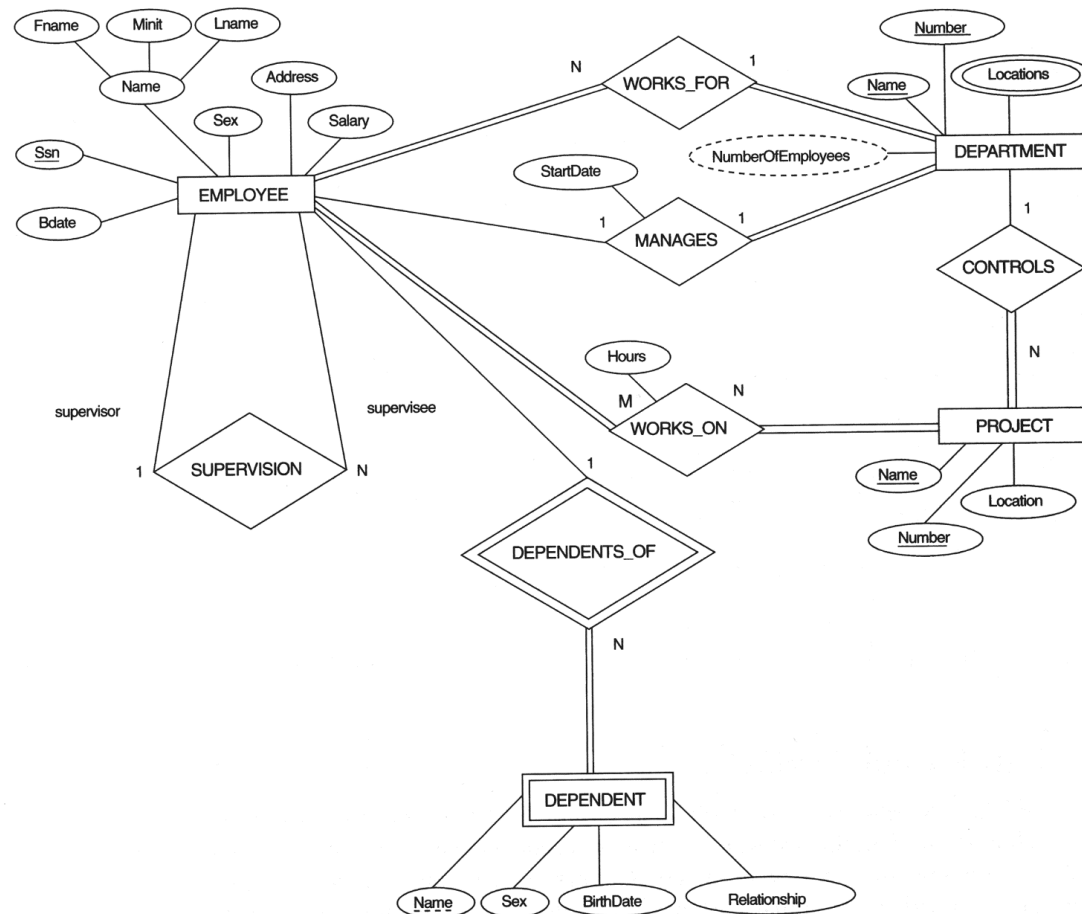
3. Logical Design

- ▶ Conceptual design is transformed into logical design
- ▶ Logical design can be expressed in a **data model**
- ▶ Example: Relational Data Model

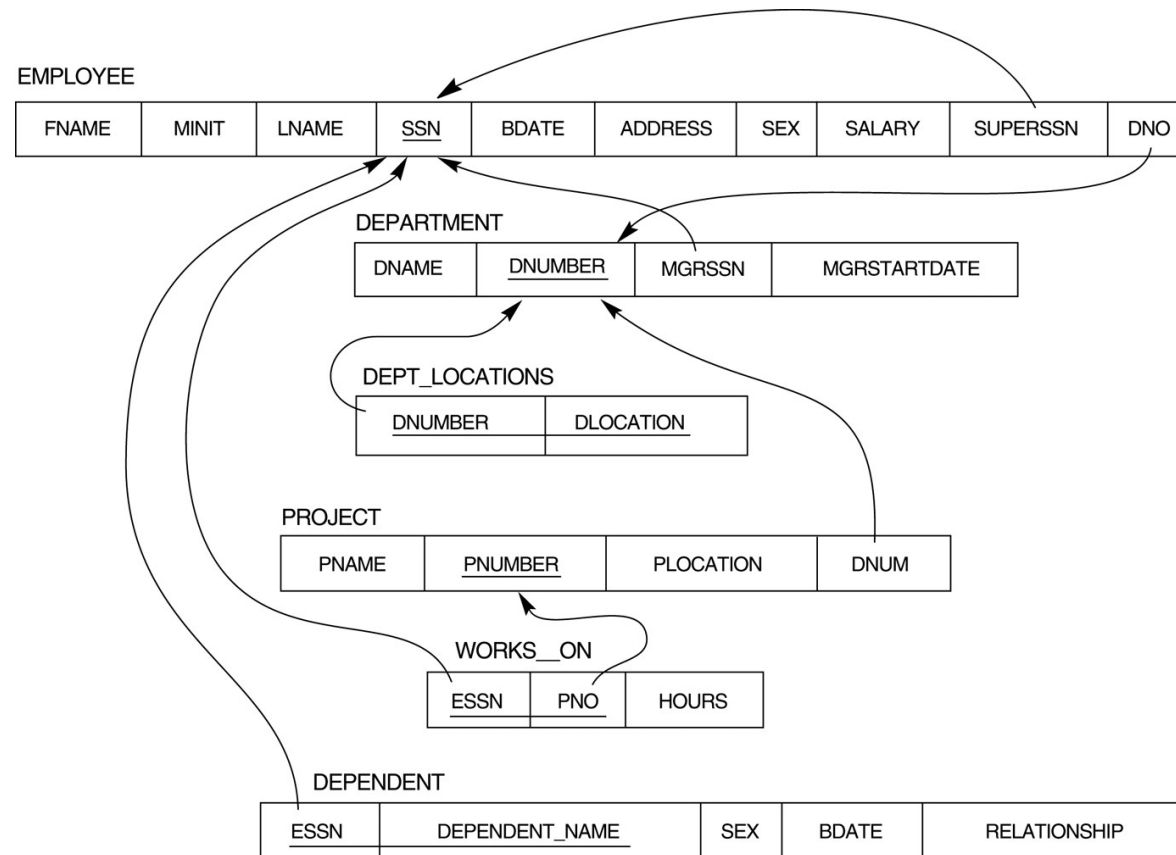
4. Physical Design

- ▶ In this stage further specifications are provided for storing and accessing the database.
- ▶ Database design is implemented , populated with actual data and continuously maintained to reflect the actual state of miniworld.

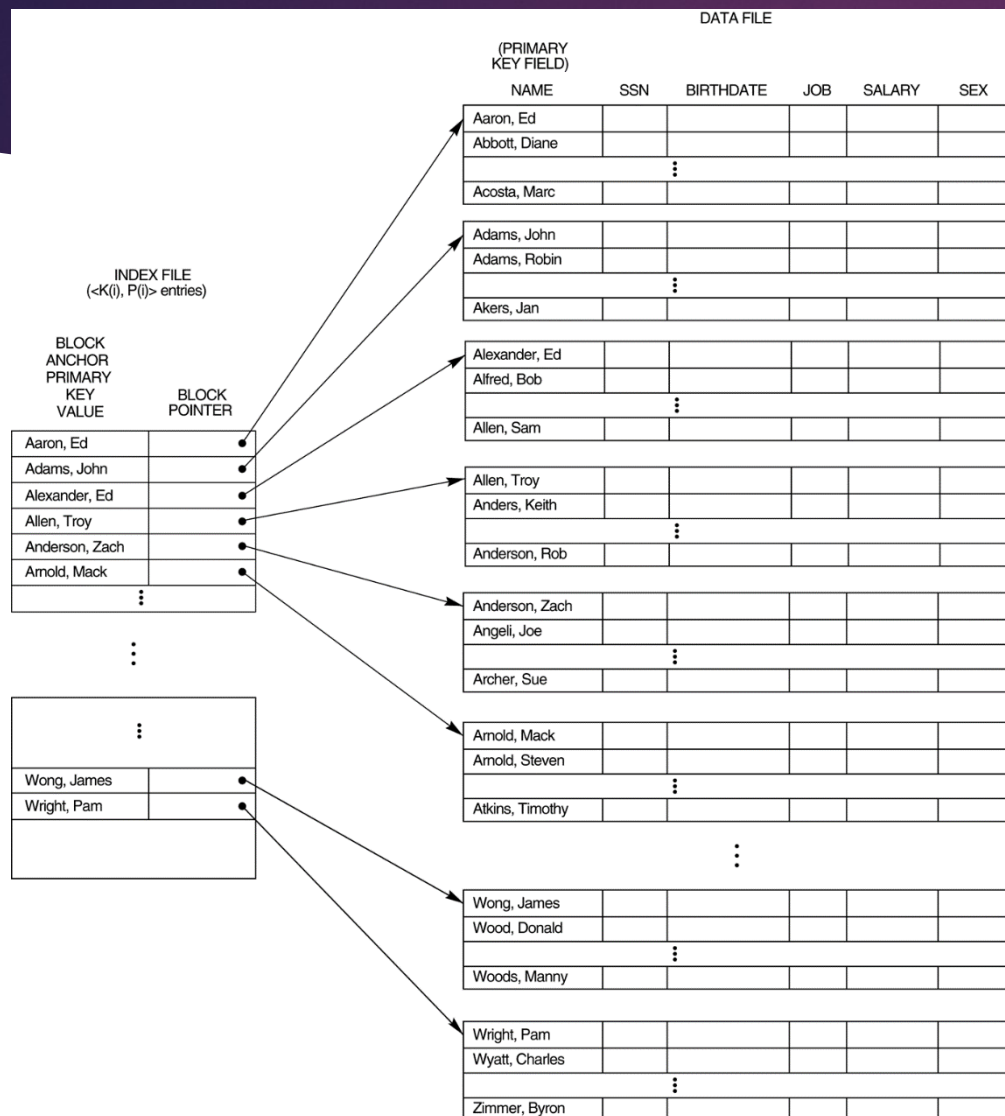
Conceptual data model – ER Diagram



Logical data model – Relational model



Physical Model



Strong Entity vs Weak Entity

- ▶ An entity is strong if it can independently exist
- ▶ An entity is weak when its existence depends on another entity.

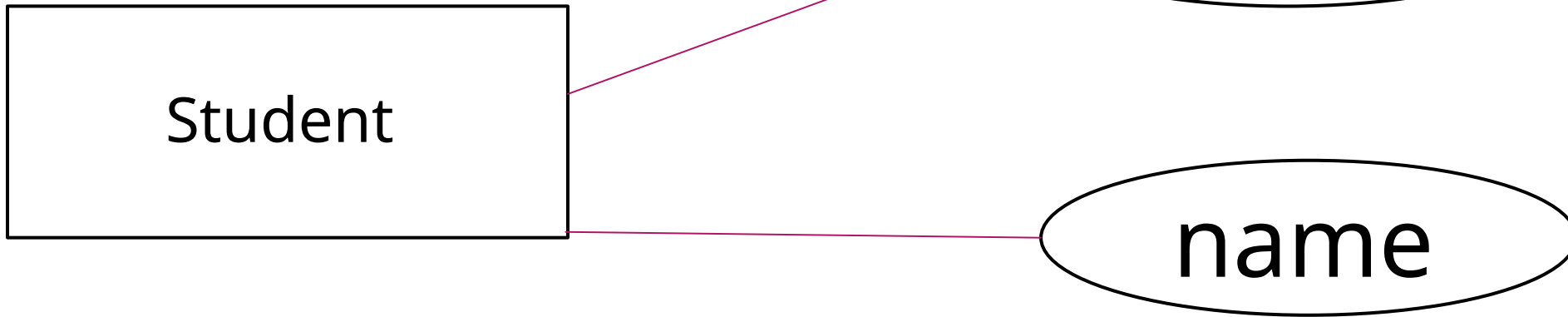
Example of weak entity

- ▶ Child can't exist without parent
- ▶ Section can't exist without class
- ▶ Account can't exist without bank
- ▶ Representation



Attributes representation in ERD (Oval)

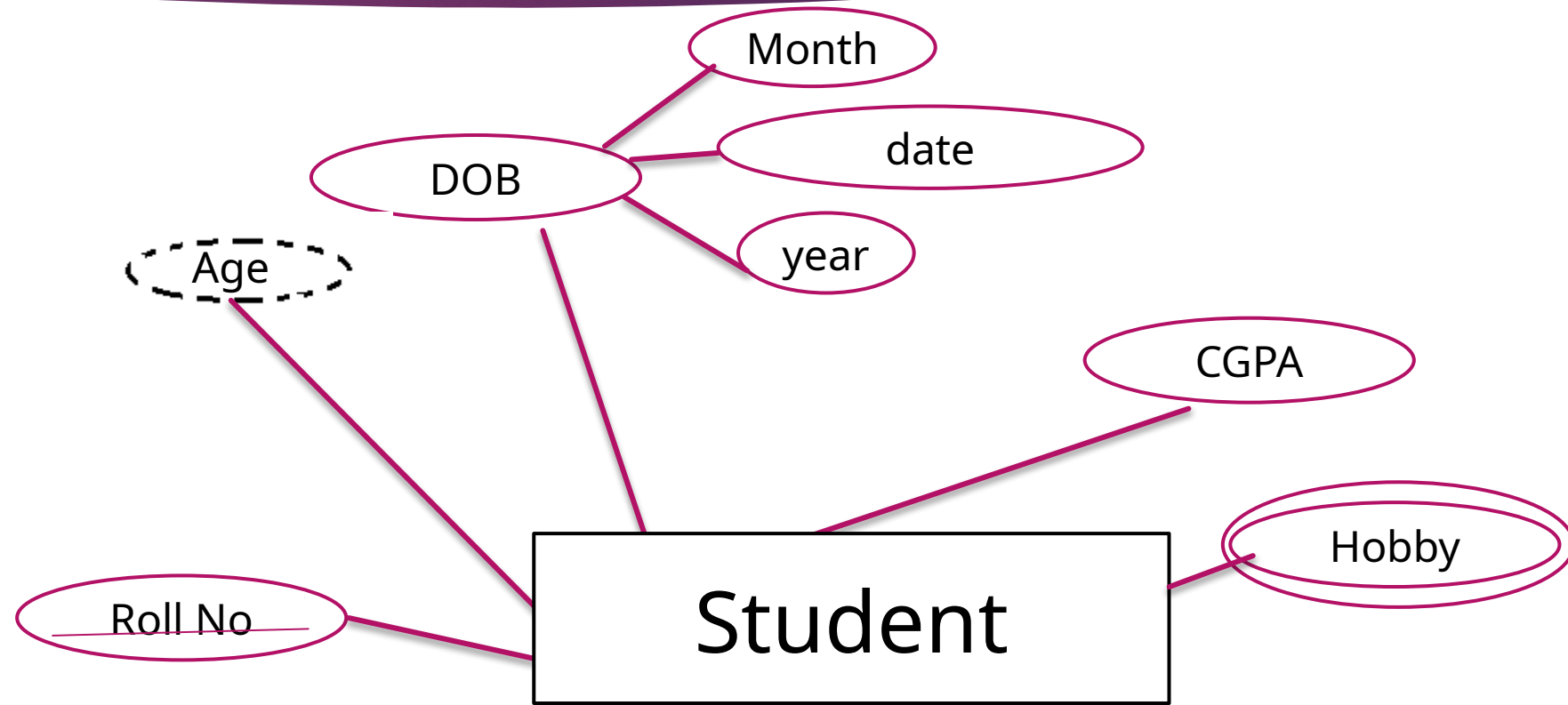
- Properties of an entity



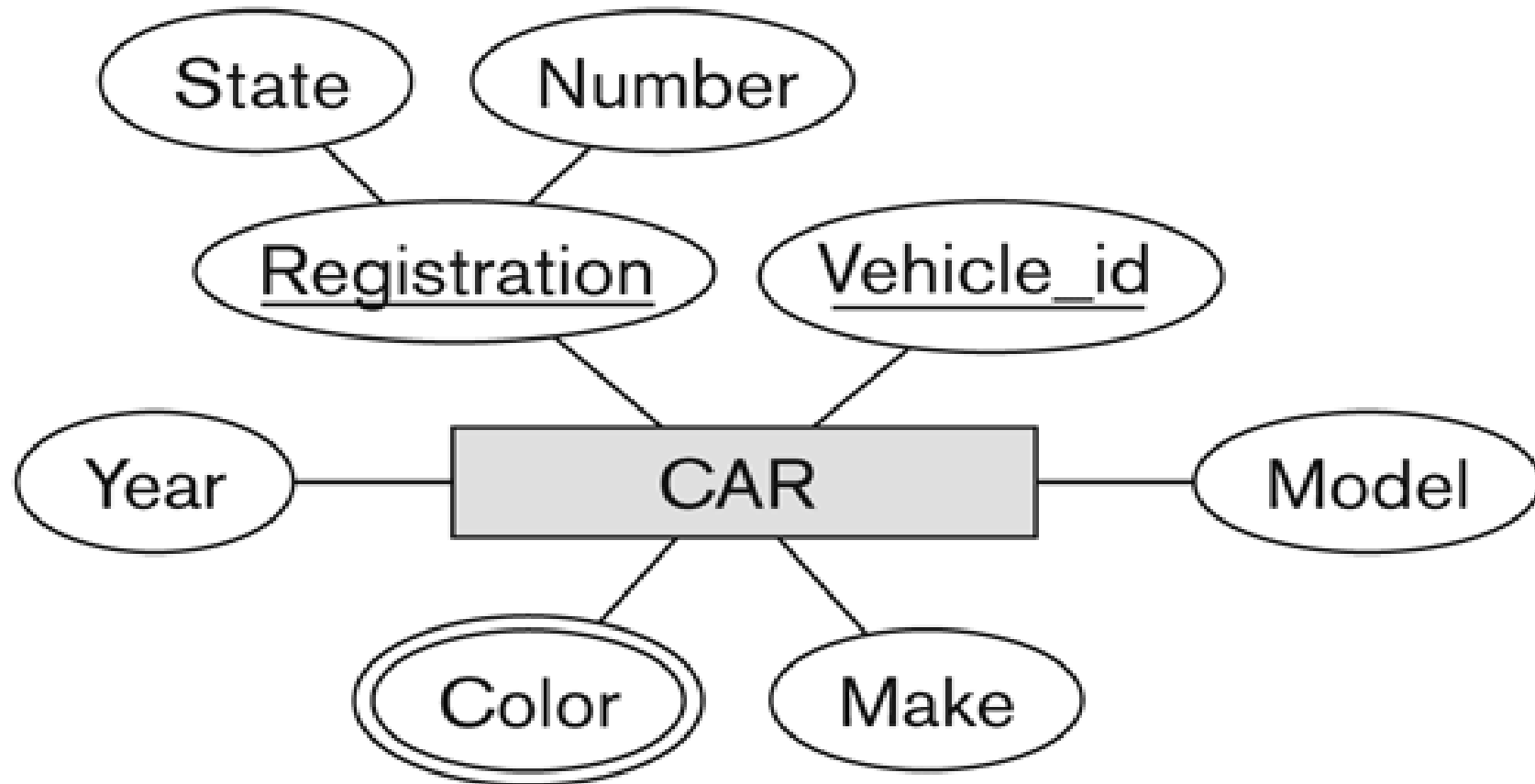
Types of an attribute

- ▶ Types of attributes
 - ▶ Key attribute
 - ▶ Composite attribute
 - ▶ Derived attribute
 - ▶ Simple attribute
 - ▶ Multivalued attribute

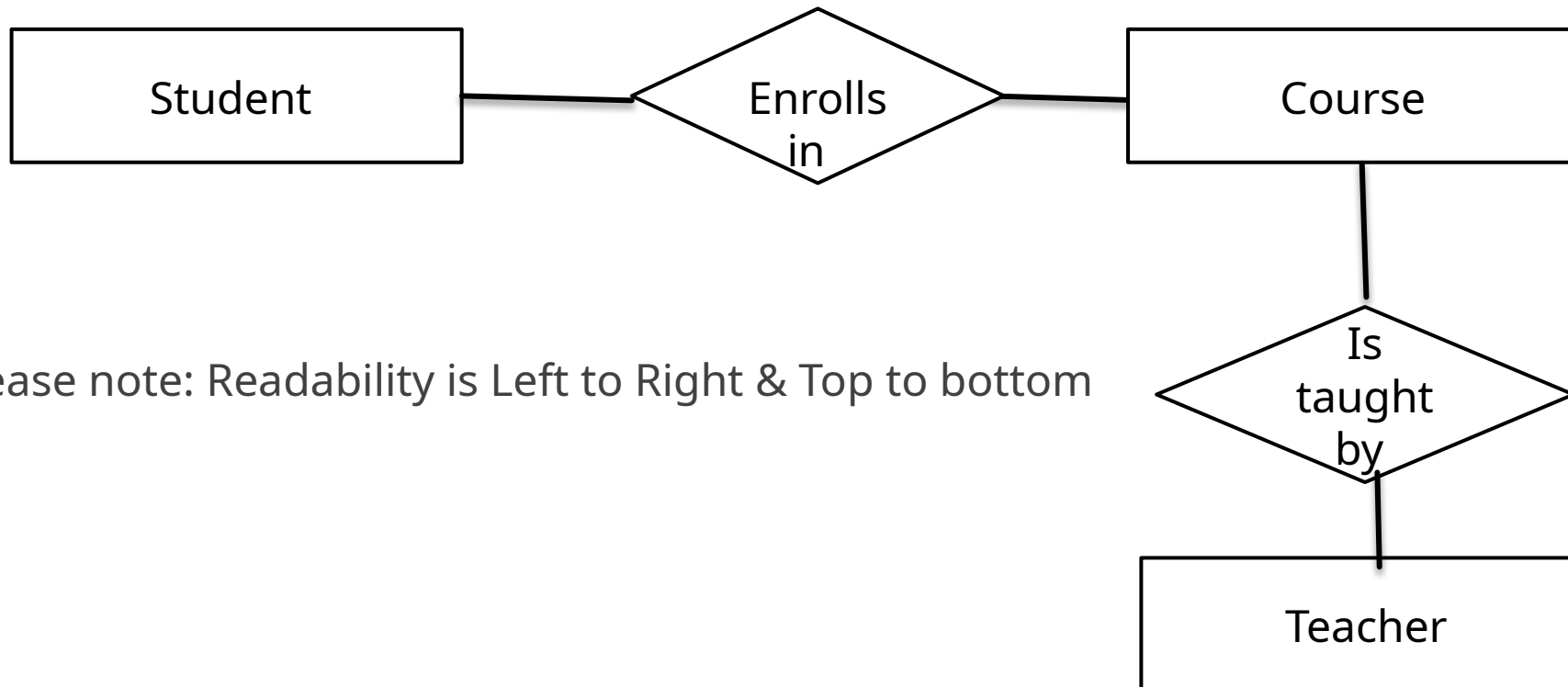
Attributes



Types of keys

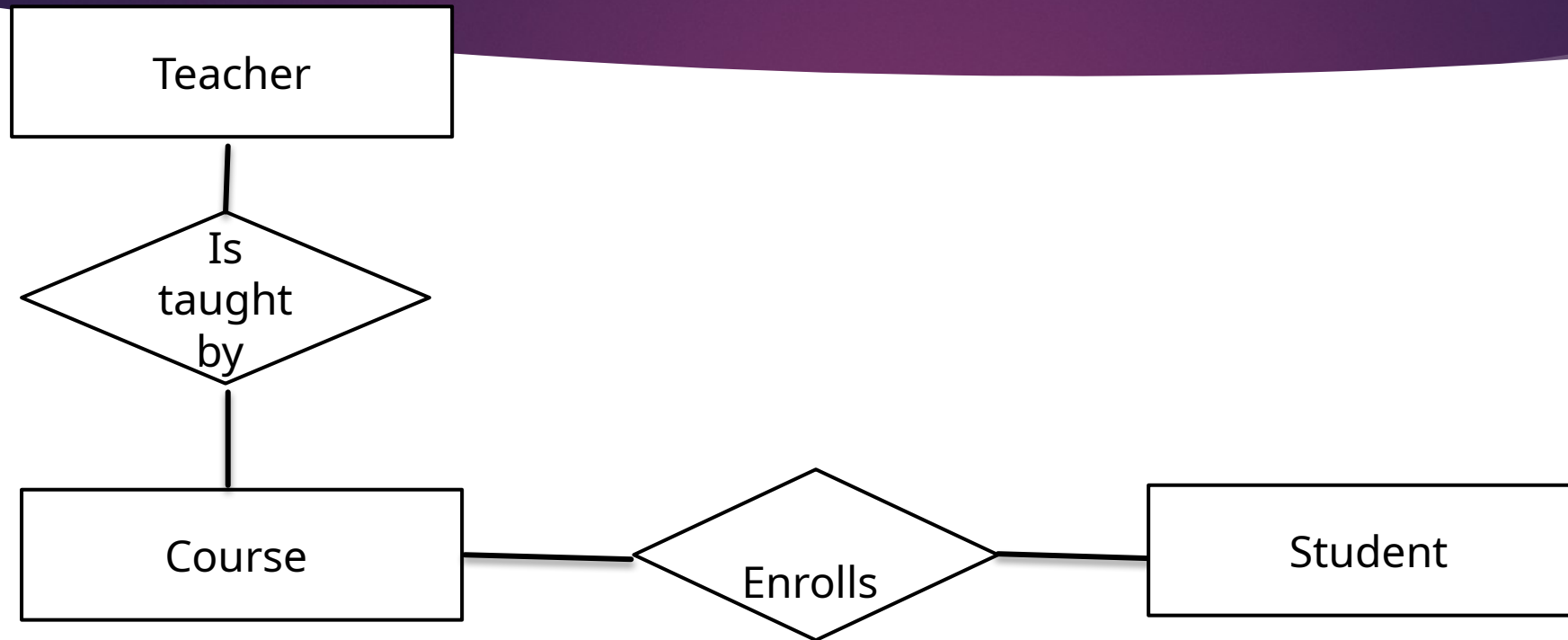


Relationship representation in ERD (Diamond)



- Please note: Readability is Left to Right & Top to bottom

Wrong structure



- ▶ Please note: Readability is Left to Right & Top to bottom
- ▶ Correct it.

Constraints on Relationship

Cardinality

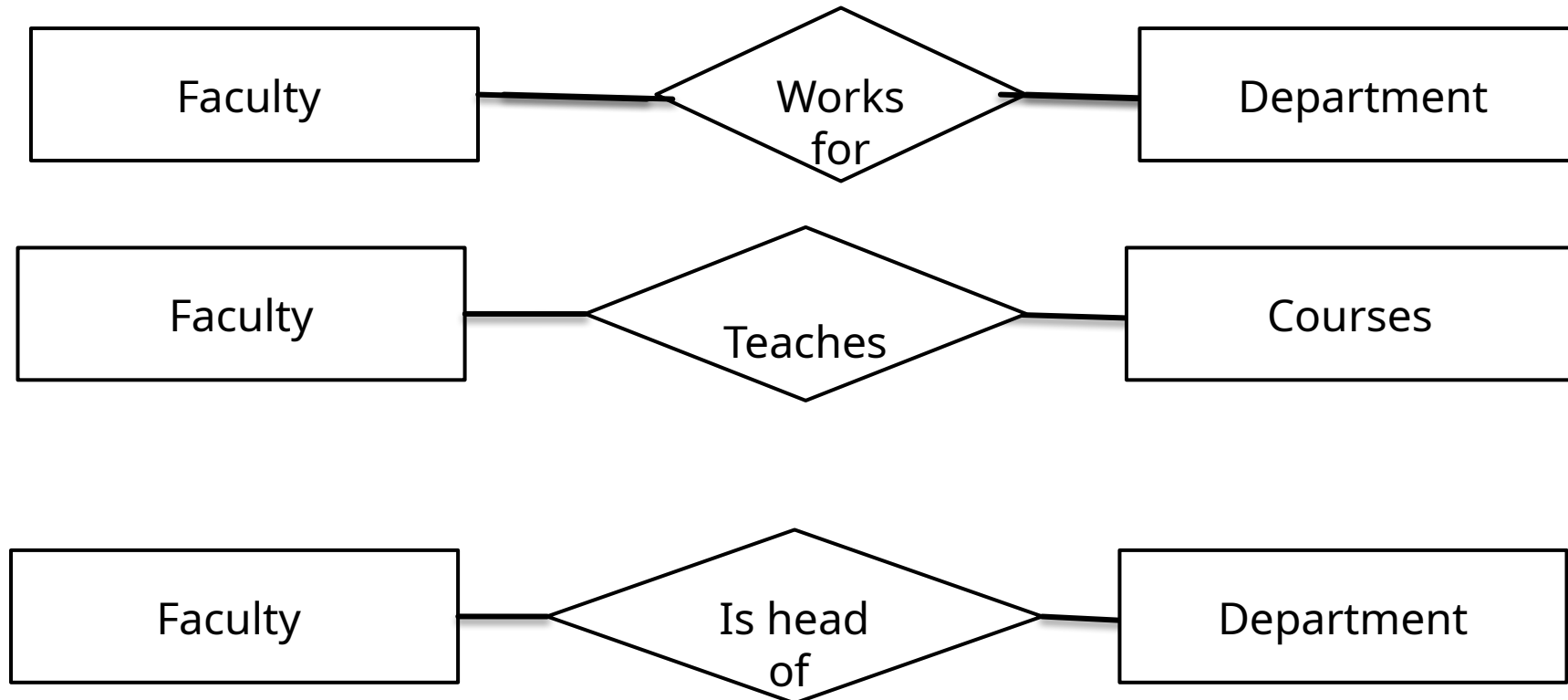
- ▶ One to many (1:N)
- ▶ One to one (1:1)
- ▶ Many to many (M:N)

Participation

- ▶ Total Participation
- ▶ Partial participation

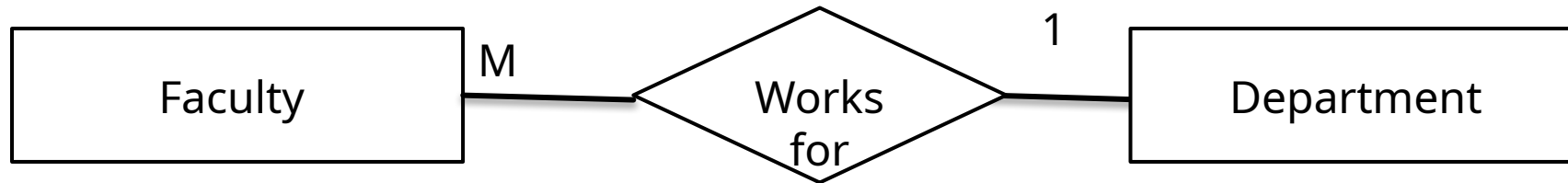


Relationship: cardinality

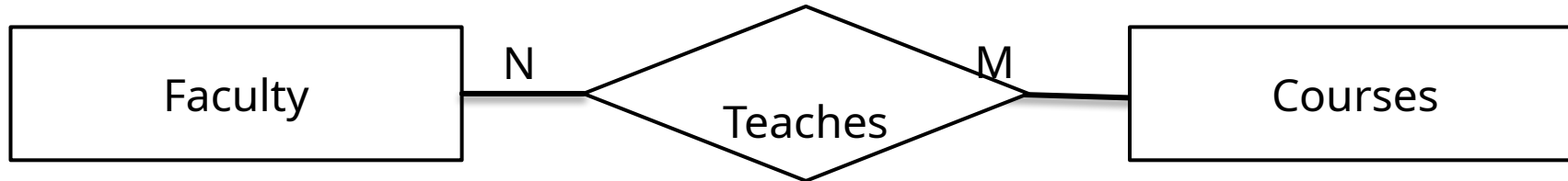


Relationship: cardinality

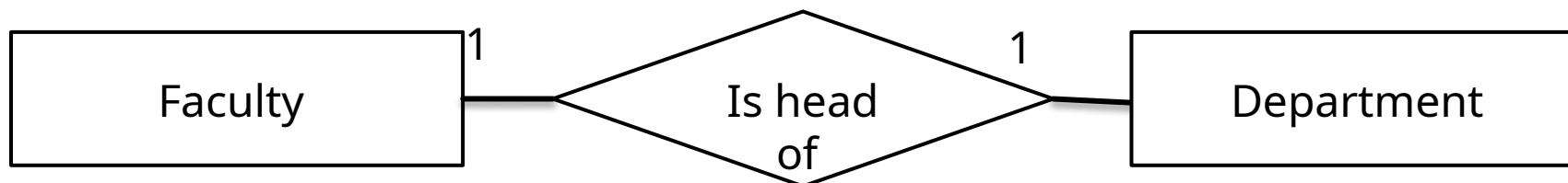
► 1 : M



► M : N

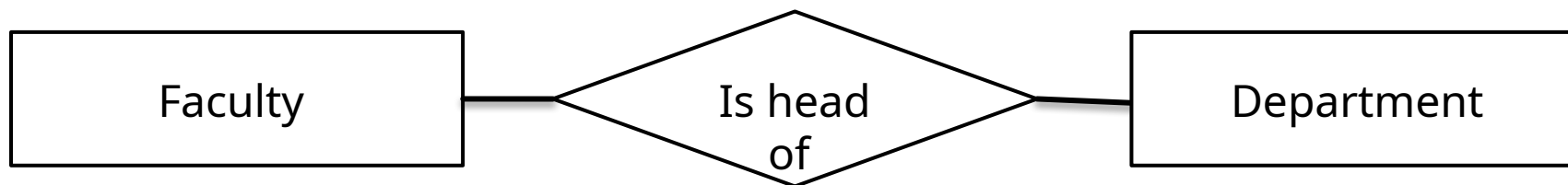
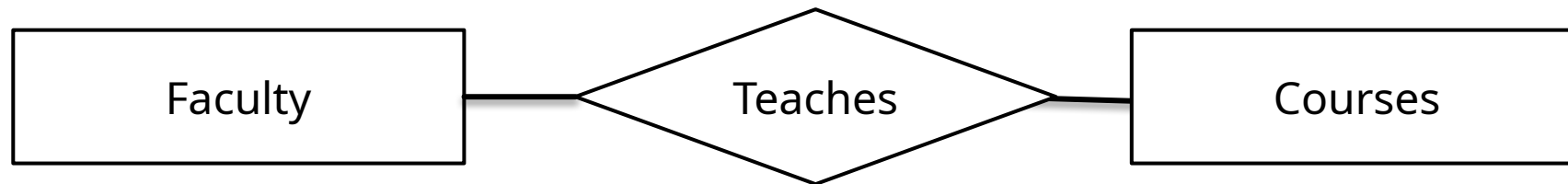
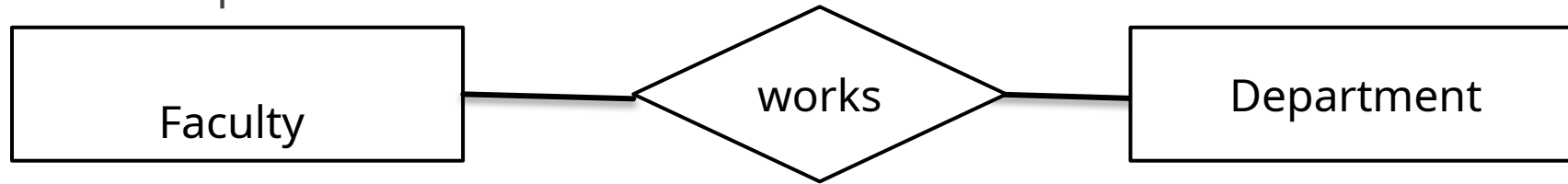


► 1 : 1



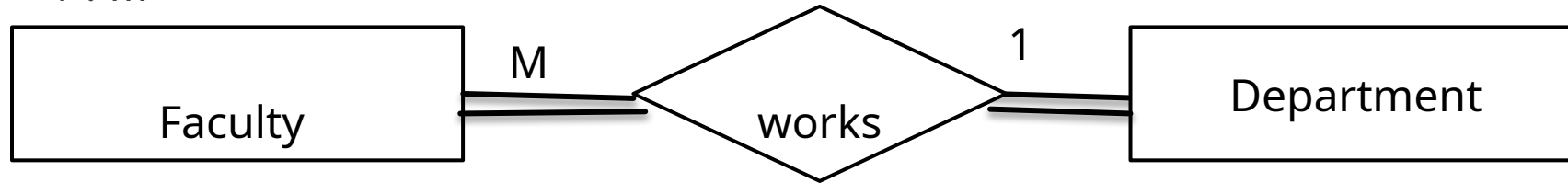
Relationship: Particitaion

► Total or partial

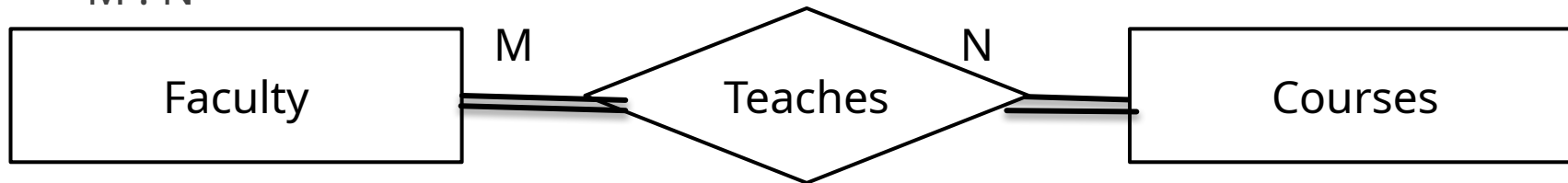


Relationship: cardinality and Participation

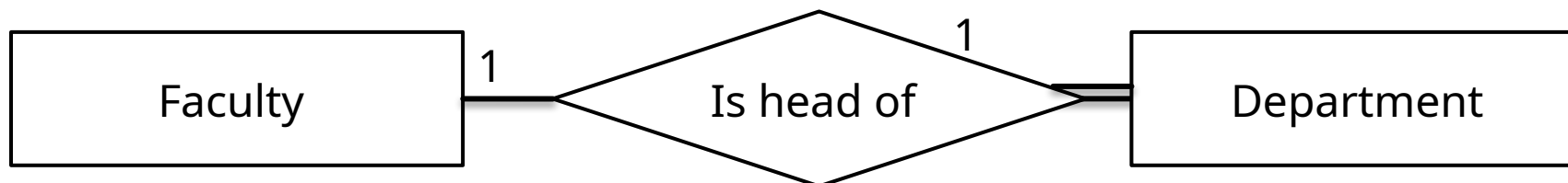
► 1 : M



► M : N

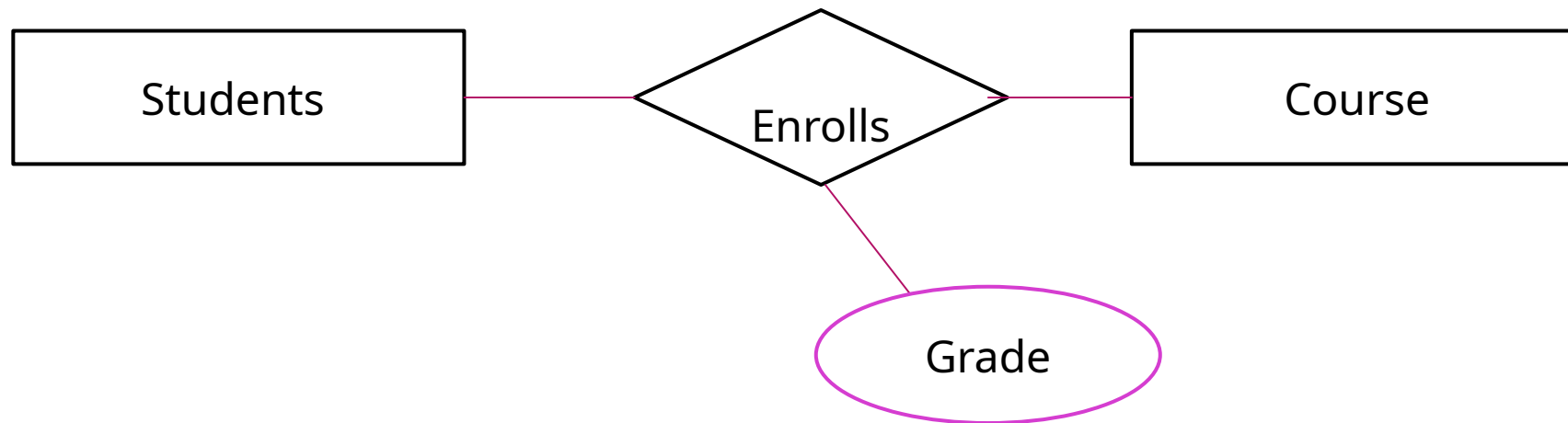


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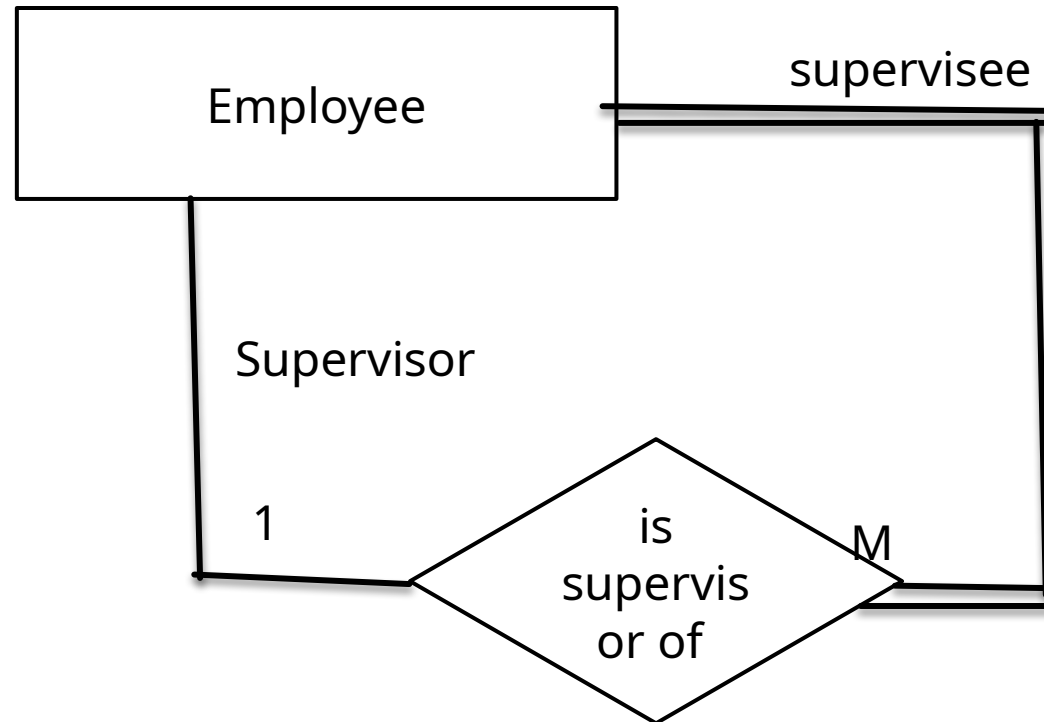
Attributes of relationship

- ▶ A relationship can have attributes:
 - ▶ For example, grade of enrolls
 - ▶ Its value for each relationship instance describes the grade a student takes when he is enrolled in a course.
 - ▶ A value of Grade depends on a particular (Student, Course) combination

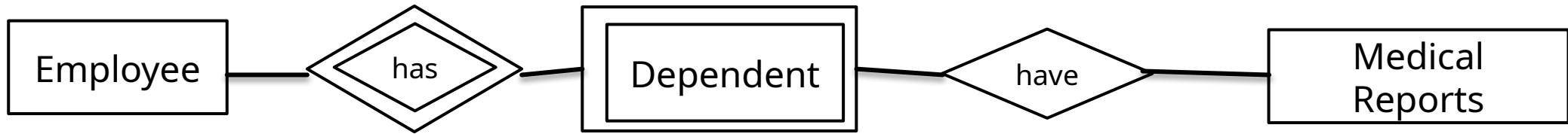


Recursive Relationship: Rolenames

- Employee is supervisor of other employees



Identifying relationship



- ▶ Relationship between a strong and weak entity is identifying.
- ▶ Why it would be needed?

Ternary relationship

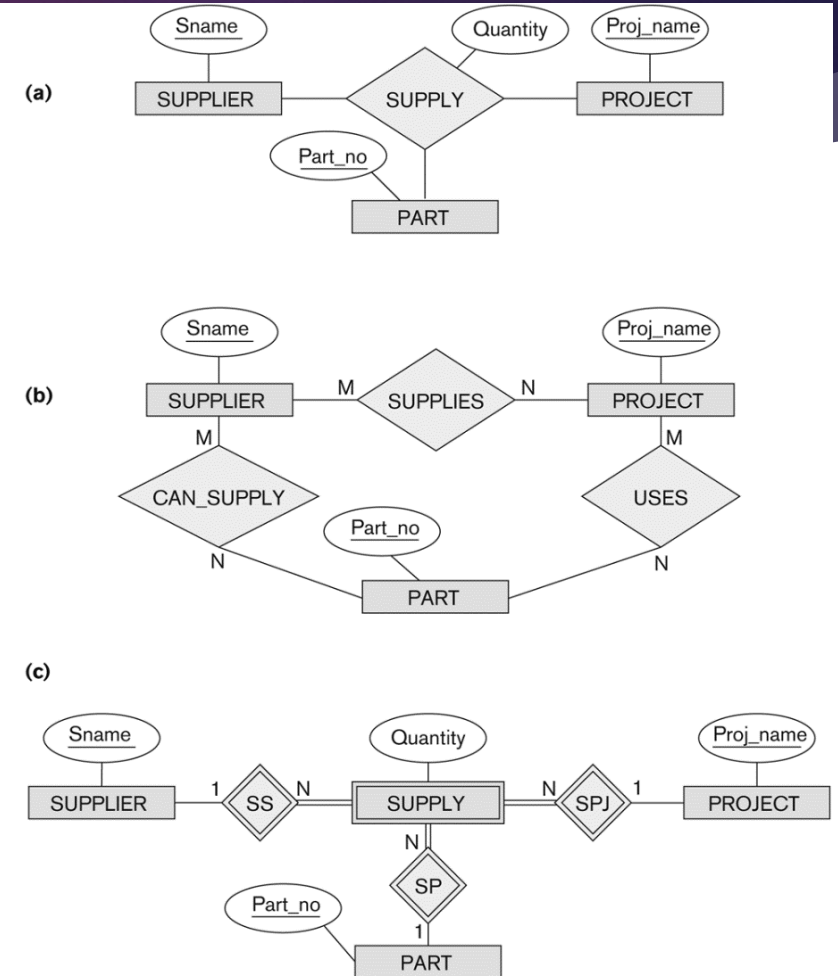
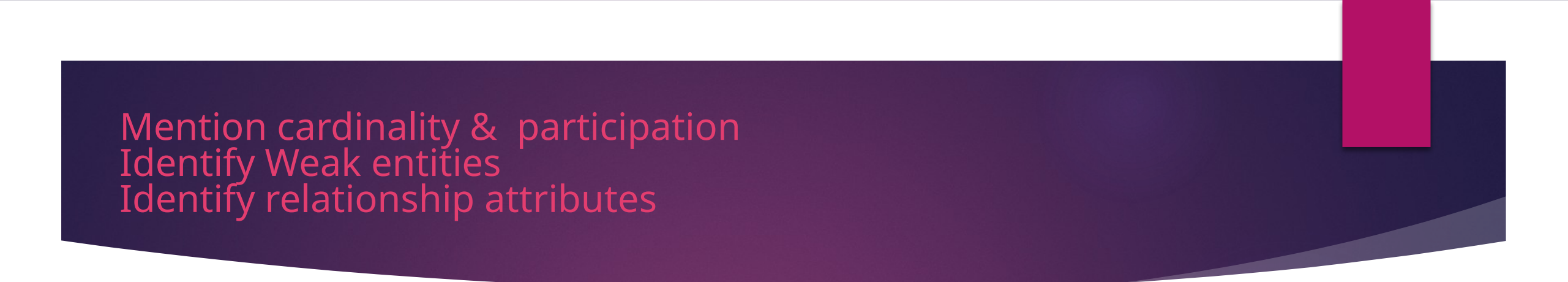


Figure 3.17

Ternary relationship types. (a) The **SUPPLY** relationship. (b) Three binary relationships not equivalent to **SUPPLY**. (c) **SUPPLY** represented as a weak entity type.

Activity: Design an ERD

- ▶ The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- ▶ A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- ▶ We store each employee's name, Social Security number, 2 address, salary, gender, and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- ▶ We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee.



Mention cardinality & participation
Identify Weak entities
Identify relationship attributes

- ▶ The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
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Self study topics

USERS OF DATABASE SYSTEM
WHEN NOT TO USE A DB

Task 1: Actors and their types involved in DBMS

- ▶ **Actors on the Scene**
 - ▶ Database administrators
 - ▶ Database Designers
 - ▶ End-users
 - ▶ Casual
 - ▶ Naïve or Parametric
 - ▶ Sophisticated
 - ▶ Stand-alone
- ▶ **Workers Behind the Scene**
 - ▶ Tool developers
 - ▶ Operators and maintenance personnel

Task 2: When not to use a DBMS

- ▶ High initial investment
- ▶ DBMS may be unnecessary
- ▶ Performance Overhead
- ▶ Single-user Applications
- ▶ Short-lived Applications

Database Users

Users may be divided

- 1. Actors on the Scene :**

Those who actually use and control the content

- 2. Workers Behind the Scene:**

Associated with the design, development, and operation of the DBMS *software and system environment*. These persons are typically not interested in the database content itself.

Database Users

Actors on the scene

- **Database administrators:** responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software, and hardware resources, controlling its use and monitoring efficiency of operations.
- **Database Designers:** responsible to define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.
- **End-users:** they use the data for queries, reports and some of them actually update the database content.

Categories of End-users

- ▶ **Casual** : access database occasionally when needed
- ▶ **Naïve or Parametric** : they make up a large section of the end-user population. They use previously well-defined functions in the form of “canned transactions” against the database. Examples are bank-tellers or reservation clerks who do this activity for an entire shift of operations.

Categories of End-users

- **Sophisticated** : these include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities. Many use tools in the form of software packages that work closely with the stored database.
- **Stand-alone** : mostly maintain personal databases using ready-to-use packaged applications. An example is a tax program user that creates his or her own internal database.

Categories of End-users

- **Application Programmers/ Software Engineers**
 - ▶ Determine the requirements of end users, especially naive and parametric end users, and develop specifications for standard canned transactions that meet these requirements.
 - ▶ **Application programmers** implement these specifications as programs; then they test, debug, document, and maintain these canned transactions.

Workers Behind the Scene

- ▶ **DBMS system designers and implementers**
 - ▶ Design and implement the DBMS modules and interfaces as a software package.
- ▶ **Tool developers**
 - ▶ Design and implement **tools**—the software packages that facilitate database modeling and design, database system design, and improved performance.
- ▶ **Operators and maintenance personnel**
 - ▶ **They** are responsible for the actual running and maintenance of the hardware and software environment for the database system.

When not to use a DBMS

- ▶ **Main inhibitors (costs) of using a DBMS:**
 - High initial investment and possible need for additional hardware.
- ▶ **When a DBMS may be unnecessary:**
 - If the database and applications are simple, well defined, and not expected to change.
 - If access to data by multiple users is not required.